

Discrete Mathematics Spring 2026

Homework 4: Proofs

Due Sunday February 22 @11:59:00pm Eastern

Show all your work to receive full credit

1. Direct proof

Using a direct proof, prove that:

- (a) The negative of an even integer is even.
- (b) The sum of any three consecutive integers is divisible by 3.

2. Proof by contradiction

Use proof by contradiction to show that:

There exist no integers a and b for which $18a + 6b = 1$.

3. Proof of if and only if (IFF)

Let x be a real number. Let *floor* of x , denoted by $\lfloor x \rfloor$, be the greatest integer less than or equal to x , and let *ceiling* of x , denoted by $\lceil x \rceil$, be the smallest integer greater than or equal to x .

For example, $\lfloor 2.13 \rfloor = 2$ and $\lceil 2.13 \rceil = 3$

Prove that

$\lfloor x \rfloor = \lceil x \rceil$ if and only if x is an integer.

4. Proof by cases

Prove that for any integers a , b , if a is even or b is even then the product ab is even.

5. Proof by contrapositive

Let $n \in \mathbb{Z}$. Use proof by contrapositive to prove the following statement:
If $8 \nmid n^3$, then n is odd.

6. Proof by contrapositive

Use proof by contrapositive to prove the following statement:

If $\log_2(3)$ is rational, then there exist positive integers m and n such that $2^m = 3^n$.

You may use the following properties of logarithms (for $a, b > 0$ and $a \neq 1$):

- **Definition:** $\log_a(b) = c$ if and only if $a^c = b$
- **Product Rule:** $\log_a(xy) = \log_a(x) + \log_a(y)$
- **Quotient Rule:** $\log_a(x/y) = \log_a(x) - \log_a(y)$
- **Power Rule:** $\log_a(x^r) = r \log_a(x)$
- **Change of Base:** $\log_a(b) = \frac{\log_c(b)}{\log_c(a)}$ for any valid base c

